

MAT 213 Brief Calculus

Name: _____

Mock Midterm D

Date: _____

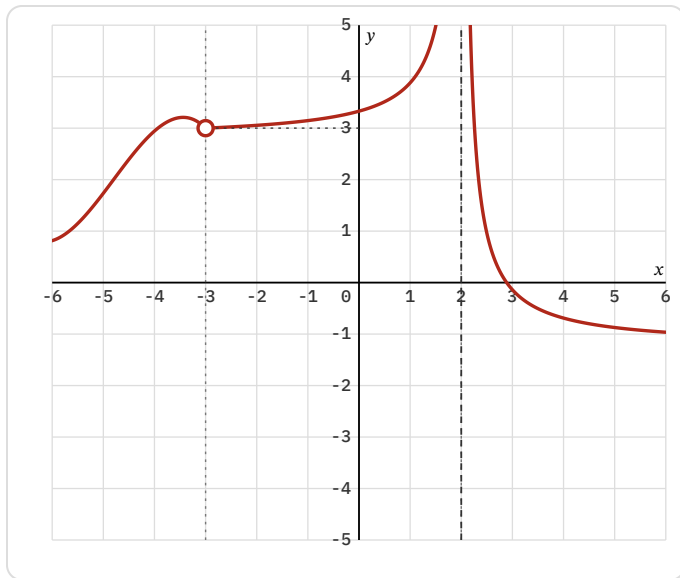
Practice midterm (AI-Free) · Sections 1.1–1.4

Score: _____ / 90

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1.

Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write **DNE**.



a.) $f(-3)$ _____ (1)

b.) $\lim_{x \rightarrow -3^-} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow -3^+} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow -3} f(x)$ _____ (1)

e.) $f(2)$ _____ (1)

f.) $\lim_{x \rightarrow 2^-} f(x)$ _____ (1)

g.) $\lim_{x \rightarrow 2^+} f(x)$ _____ (1)

h.) $\lim_{x \rightarrow 2} f(x)$ _____ (1)

8 pts

2. Let $f(x) = \frac{1}{(x-3)^2}$. Answer with a number, ∞ , $-\infty$, or DNE.

5 pts

a.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

c.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

d.) $\lim_{x \rightarrow \infty} f(x)$ _____ (1)

e.) The vertical asymptote is $x =$ _____ (1)

3. Calculate each limit algebraically.

4 pts

a.) $\lim_{x \rightarrow 3} (-3x^2 + 6x - 2)$ (1)

b.) $\lim_{x \rightarrow -1} \frac{x^2 - x - 4}{x - 3}$ (1)

c.) $\lim_{x \rightarrow 4} \sqrt{x^2 + 20}$ (1)

d.) $\lim_{x \rightarrow 7} 9$ _____ (1)

4.

Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE.

a.) $\lim_{x \rightarrow 4} \frac{x-4}{\sqrt{x}-2}$ _____ (2)

b.) $\lim_{x \rightarrow 0} \frac{-6x^2-5x}{x}$ _____ (2)

4 pts

5.

Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results.

6 pts

a.) $\lim_{x \rightarrow \infty} \frac{5x^2-9x}{6x}$ _____ (2)

b.) $\lim_{x \rightarrow -\infty} \frac{2}{5x^2-8x+7}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{5x^3+9x^2-5x}{-5x^3-6}$ _____ (2)

6. Find each limit (end behavior). Use ∞ or $-\infty$ when the outputs grow without bound. 4 pts

a.) $\lim_{x \rightarrow \infty} (-4x^5 - 6x^4 - 6x^2 - 5)$ _____ (1)

b.) $\lim_{x \rightarrow \infty} (-1 - 3e^{-x})$ _____ (1)

c.) $\lim_{x \rightarrow -\infty} (-6 - \frac{8}{x^2})$ _____ (1)

d.) $\lim_{x \rightarrow -\infty} (x^3 - 5x^2 - 4x)$ _____ (1)

7. Let 6 pts

$$f(x) = \begin{cases} \frac{x^2 - 4x + 3}{x - 3} & x < 3 \\ 4x - 10 & x > 3 \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 6} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow 0} f(x)$ _____ (1)

c.) $f(3)$ _____ (1)

d.) $\lim_{x \rightarrow 3^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow 3^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow 3} f(x)$ _____ (1)

8.

Is $f(x)$ continuous at $x = -2$?

5 pts

$$f(x) = \begin{cases} \frac{x^2 + x - 2}{x + 2} & x \neq -2 \\ -3 & x = -2 \end{cases}$$

a.) $f(-2)$ _____ (2)

b.) $\lim_{x \rightarrow -2} f(x)$ _____ (2)

c.) Is f continuous at $x = -2$? (1)

(A) Yes — $f(-2)$ exists, the limit exists, and they are equal

(B) No — the one-sided limits differ, so the limit does not exist

(C) No — the limit exists but does not equal $f(-2)$ (D) No — $f(-2)$ is undefined

9.

Classify each function at the given x -value.

3 pts

a.) $f(x) = \frac{x^2-25}{x+5}$ at $x = -5$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

b.) $f(x) = \frac{x-2}{x^2-3x+2}$ at $x = 1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

c.) $f(x) = \frac{x+1}{x^2-2x-3}$ at $x = -1$ (1)

- (A) Continuous
- (B) Removable discontinuity
- (C) Jump discontinuity
- (D) Infinite discontinuity

10.

Let $V(t) = t^2 - 30t + 600$, where $V(t)$ represents the number of gallons of water in a tank t minutes after a drain is opened. Find the average rate of change over $[8, 11]$.

a.) $V(8)$ _____ (2)

b.) $V(11)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

d.) Which sentence correctly states the result **with units**? (1)

(A) The amount of water in the tank changes by a total of -11 gallons from $t = 8$ to $t = 11$.

(B) At exactly $t = 11$ minutes, the amount of water in the tank is changing at a rate of -11 gallons per minute.

(C) Between $t = 8$ and $t = 11$, the amount of water in the tank changes at an average rate of -11 minutes per gallon.

(D) Between $t = 8$ and $t = 11$ minutes, the amount of water in the tank changes at an average rate of -11 gallons per minute.

7 pts

11.

Let $f(x) = \frac{6}{x}$. Calculate the average rate of change of f over each interval. (Rounded answers to 2+ decimal places are fine.)

a.) $[3, 5]$ _____ (1)

b.) $[3, 4]$ _____ (1)

c.) $[3, 3.1]$ _____ (1)

d.) $[3, 3.01]$ _____ (1)

e.) As the interval shrinks toward $x = 3$, these values approach... (1)

(A) $\frac{2}{3}$

(B) $-\frac{4}{3}$

(C) $-\frac{2}{3}$

(D) $\frac{1}{3}$

5 pts

12.

Let $f(x) = -x^2 - 3x - 5$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table.

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = -3, h = 1$ _____ (1)

c.) $x = -3, h = 0.1$ _____ (1)

d.) $x = -3, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

7 pts

13.

Let $f(x) = \frac{2}{x}$. Use the **limit definition** $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ to find $f'(x)$.

a.) $f'(x) =$ _____ (3)

b.) $f'(-2) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 2$ _____ (2)

d.) What does $f'(-2) = -\frac{1}{2}$ mean? (1)

(A) The average rate of change of f from $x = 0$ to $x = -2$ is $-\frac{1}{2}$.

(B) The value of f at $x = -2$ is $-\frac{1}{2}$.

(C) The instantaneous rate of change of f at $x = -2$ is $-\frac{1}{2}$ — the slope of the tangent line there.

(D) The slope of the secant line from $x = -2$ to $x = -2 + 1$ is $-\frac{1}{2}$.

8 pts

14.

Let $f(x) = 2x^2 + 5x + 5$. Find the equation of the tangent line to the graph of f at the point $(2, 23)$.

a.) $f'(x) =$ _____ (3)

b.) Slope at $x = 2$ _____ (2)

c.) Tangent line _____ (3)

8 pts

15. Let $f(x) = x^2 - 3$. Complete the limit definition, then interpret $f'(3)$. 6 pts

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{((x+h)^2 - 3) - (x^2 - 3)}{h} = ?$$

a.) $f'(x) =$ _____ (3)

b.) $f'(3)$ _____ (2)

c.) What does $f'(3) = 6$ tell you? (1)

(A) The slope of the tangent line to the graph of f at $x = 3$ is 6.

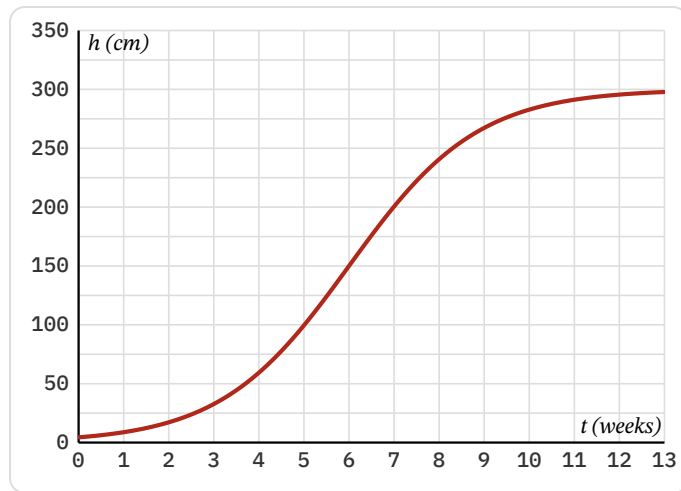
(B) The slope of the secant line from $x = 0$ to $x = 3$ is 6.

(C) The graph of f crosses the y -axis at 6.

(D) The point $(3, 6)$ is on the graph of f .

16.

A sunflower seedling is planted and its height h (in centimeters) after t weeks is graphed below. Decide whether each statement is true or false.



- a.) T F The slopes of the tangent lines get steeper up to $t = 3$, and then the slopes start to flatten out. (1)
- b.) T F The average rate of change over $[0, 12]$ is greater than the instantaneous rate of change at $t = 7$. (1)
- c.) T F The seedling grows faster (instantaneous rate) at week 5 than at week 1. (1)
- d.) T F The average rate of change over $[7, 9]$ is less than over $[2, 4]$. (1)

4 pts
